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CSE437: DATA SCIENCE: CODING WITH REAL-WORLD DATA

Quiz #3

Student ID: **Date**: September 7, 2026

Name: **Marks**: 20

Time: 40 min

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Consider the following 1D dataset: **{2, 3, 5, 8, 20}**. We want to cluster this data into **K=2** clusters.

1. Consider doing **K-Medoids** clustering for the given dataset.
   Initial medoids are: $M_1 = 3 \; and \; M_2 = 8$.
   a. Calculate the **total cost** (sum of distances from each point to its assigned medoids) for the initial medoid configuration. [3]
   b. Now consider swapping medoid M_2 from 8 to 5. Calculate the new total cost with medoids $M_1 = 3 \; and \; M_2 = 5$. Would the algorithm accept this swap? Why or why not? [5]
   c. The value 20 is an outlier. Explain which algorithm (K-Means or K-Medoids) would be more robust to this outlier and why. Support your answer with the reference to how each algorithm updates its cluster representatives. [2]

2. Perform **K-Means clustering** using Euclidean distance until convergence for the same dataset. Show the cluster assignments and updated centroids at each iteration. Initial centroids are randomly chosen as: $\mu_1 = 3$ *and* $\mu_2 = 8$. [10]